



Research paper

Cooperative game method of heterogeneous unmanned surface vehicles based on distributed decision-making framework

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ABSTRACT

Due to targeted design, unmanned surface vehicles (USVs) with different functions often show significant heterogeneous characteristics. This heterogeneity allows the USV cluster to perform more complex and diverse tasks. This paper innovatively proposes a cooperative game method for a heterogeneous USV cluster based on a distributed decision-making framework, which aims to significantly improve its decision-making quality. Specifically, we first combine the underactuated characteristics of the USV and the actual mission requirements and use the geometric method to accurately calculate the position of the guidance point. On this basis, we introduce the concept of 'neighbor USV', and construct a distributed decision-making framework based on neighbor USV. The framework further enhances the coordination ability between USVs through a hierarchical decision-making mechanism. In the hierarchical decision-making system, the upper decision-making is mainly responsible for solving the sub-task allocation problem of each USV. The lower decision-making focuses on the specific action strategy of each USV. In order to verify the feasibility and superiority of the proposed method, we carried out Monte Carlo simulation experiments. The experimental results show that this method significantly improves the coordination efficiency and task execution ability of the USV cluster while ensuring real-time decision-making. Through the research of this paper, we provide a new idea for the cooperative control of heterogeneous USV clusters and provide a reference for the broad application of USV clusters in the future complex task environment.

1. Introduction

1.1. Motivation and incitement

In recent years, with the rapid progress of hardware, software, and artificial intelligence technology, unmanned vehicles have played an essential role in many fields (Mu et al., 2023; Feng et al., 2024; Tian et al., 2024). In the marine environment, USVs with high autonomy and powerful computing capabilities have become the focus of current research (Wu et al., 2021). Due to its unique advantages, USV is widely used in marine scientific research, regional search, warning patrol, target tracking, and other tasks (Yao et al., 2023; Li et al., 2023). At present, USVs tend to be modular design. According to different mission requirements, various types of USVs are designed with different sensors and mission modules. Heterogeneous USVs containing multiple types can accomplish more complex tasks through intra-group collaboration (Liu et al., 2023).

USV cluster interception is a typical USV game scenario, which is very important for maintaining the harmony of the sea area (Rao et al., 2022). In the face of USVs close to specific boundaries, the use of USVs as

a response shows high safety and efficiency (Zhang et al., 2024b). Such interactive scenarios are rich in highly dynamic and complex features, which impose more stringent requirements on rapid decision-making capabilities (Gu et al., 2024). With the increasing advancement of unmanned technology, it has become a foreseeable challenge to simultaneously respond to multiple USVs entering a specific area (Wang et al., 2023a). Moreover, these entrants may surpass managers in sports performance, which undoubtedly adds additional difficulty to managers (Yang and Yan, 2024).

1.2. Literature review

In recent years, the multi-agent game has gradually become a research hotspot, and scholars have made significant progress in this field.

Xu Jiwei et al. established a confrontation model based on uncertain information for the problem of multi-UAV confrontation. Information uncertainty is simulated by introducing the situation matrix when establishing the revenue function. The simulation results show that the proposed strategy can perform well in the multi-UAV confrontation

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problem (Xu et al., 2020). Yan Xinghui et al. proposed a cooperative strategy using multiple inferior missiles to intercept a single highly maneuvering target. The reachability analysis is carried out by using the control boundary of the target as a priori knowledge. The quantitative advantage of the missiles can be effectively transformed into a combat advantage by covering the reachable set of the target with the reachable sets combination of the missiles (Yan et al., 2020). Liu Yanfang et al. established a non-zero sum linear quadratic differential game model for the problem of two missiles intercepting a single target. The guidance law is derived from the Nash equilibrium strategy set of the game. Simulation shows that the above strategy can improve the probability of successful interception through cooperation between interceptors (Liu et al., 2012). Eloy et al. studied the problem of team conflict between N hunters and M evaders. The classical differential game theory is extended to solve the allocation problem and the multiplayer pursuit problem at the same time. Regardless of the actual strategy implemented by the opponent, it provides guaranteed performance for each team (Garcia et al., 2021).

The above studies all adopt a centralized decision-making framework. Although the centralized decision-making method has been relatively mature, it still has many inherent defects, such as poor robustness, insufficient system scalability, and decision delay when the number of nodes is large. In addition, when the agent with command and control software cannot work, it poses a huge threat to the successful completion of the mission. **The distributed decision-making method represented by machine learning and game theory can solve the problems well.**

Zhang et al. designed the optimal control strategy for the multi-agent pursuit-evasion game problem based on the reinforcement learning method. According to the number of evaders, the problem of multiple pursuers and multiple evaders is decomposed into a more straightforward game of multiple pursuers and an evader. A data-driven reinforcement learning algorithm is developed to learn the optimal control strategy online using measurable behavior data (Zhang et al., 2024a). Wang Yuanda et al. considered a game scenario involving multiple pursuers and a better evader and proposed a distributed cooperative strategy with communication. A learning-based communication mechanism is used to solve the problem of cooperative pursuit control. Regarding sharing information, the broadcast method is not used, but a ring topology network and a leader-follower topology network are constructed to communicate (Wang et al., 2020). Li Fanbiao et al. proposed a distributed capture strategy optimization method for the pursuit-evasion game of multiple USVs. Considering the limited sensing range of each pursuer, a multi-agent near-end strategy optimization method combined with a new speed control mechanism is used to guide the pursuer to approach the escaper (Li et al., 2024). Xia Jiawei et al. proposed a hunting algorithm based on multi-agent reinforcement learning for the multi-target hunting problem of unmanned surface vehicle formation. The cooperative hunting problem of USV formation is modeled as a decentralized partially observable Markov decision process, and a distributed partially observable multi-target hunting proximal strategy optimization algorithm for USV is proposed. Simulation experiments verify the effectiveness of the above algorithm in test scenarios with different numbers of USVs (Xia et al., 2023).

Although offline task planning dominated by reinforcement learning can use the rich strategy library to achieve efficient decision-making in multi-agent collaborative tasks (Yang et al., 2024), the performance of machine learning algorithms depends largely on the quality and quantity of training data. When facing special tasks with less available training data, the online learning ability of machine learning models needs to be strengthened.

Lei Cheng et al. studied the problem of a multi-person pursuit-evasion game in an obstacle environment. In order to obtain the distributed control strategy, the communication topology graph is used to quantify the interaction between the participants and calculate the Nash equilibrium of the two sides of the game. The adaptive method is used to obtain the control parameters of the opponent, which is used to

update multiple distributed control strategies (Cheng and Yuan, 2023). Wang Yueying et al. proposed a distributed pursuit game algorithm for unmanned surface vehicles based on the threat potential field. Among them, the threat potential field is established to evaluate the threat level of the target. The concepts of energy peak and energy valley are proposed to guide strategy selection. The particle swarm optimization algorithm is used to solve the proposed game optimization model to obtain the strategy (Wang et al., 2023b). Deng Ziquan et al. studied the game problem involving a single attacker and multiple defenders and proposed a distributed pursuit-defense strategy. It is proved that the proposed strategy can ensure that the attacker is captured before entering the protected area in a limited time (Deng and Kong, 2020). Zhou et al. proposed a distributed optimal solution for the multi-agent pursuit game problem. By analyzing the optimal control strategy, the Nash equilibrium is solved by combining the Hamilton-Jacobi-Isaacs equation. Simulation and actual flight experiments have successfully verified the effectiveness of the proposed strategy (Zhou and Chen, 2024).

The above research shows that the online decision-making method under the distributed decision-making framework shows certain advantages in dealing with multi-agent games.

1.3. Contribution and paper organization

Based on a full understanding of the existing research, this paper proposes a distributed collaborative decision-making method of heterogeneous unmanned surface vehicles, which can deal with targets with various motion performances based on meeting real-time decision-making. The main innovations and contributions of this paper are as follows :

1. Given the limitations of the existing research in predicting the guidance point, **this paper introduces the reachable set theory of the underactuated system, which breaks through the limitations of the complete maneuverability assumption or the simplification of the first-order motion parameters.** The proposed prediction method fully considers the physical constraints, such as the minimum turning radius of the USV. It significantly improves the accuracy of state prediction in the high-speed pursuit-evasion scenario.
2. This paper designs a distributed decision-making framework **based on the concept of neighbor USVs** to address the computational challenges in complex game scenarios. **By considering local policy interaction, the framework effectively reduces the complexity of collaborative task modeling and improves the quality of decision-making while ensuring real-time decision-making.**
3. Aiming at the problem of dimension disaster in the traditional collaborative game method, **this paper proposes a two-layer decision-making system of task-action decoupling.** The system realizes the effective decomposition of decision complexity through the hierarchical design of the upper layer, which solves task allocation, and the lower layer, which solves action strategy. This provides a new idea for solving the USVs' cooperative game problem.
4. This paper abandons the traditional fixed-scene test method for simulation tests and constructs a systematic evaluation framework based on a Monte Carlo experiment. By comprehensively sampling the main influencing factors, the framework objectively verifies the algorithm's robustness and superiority in various scenarios and provides a more scientific method for evaluating its performance.

The organizational structure of this paper is arranged as follows: **Section 2** first establishes the mathematical model of the study and proposes the fundamental assumptions. **Section 3** provides a mathematical description of the reachable set by considering the underactuated motion characteristics, and further proposes a distributed game framework based on a hierarchical decision-making system. **Section 4** employs the Monte Carlo simulation method to systematically validate the proposed cooperative game approach from three dimensions: robustness, superiority, and scalability. Furthermore, the potential of the proposed method

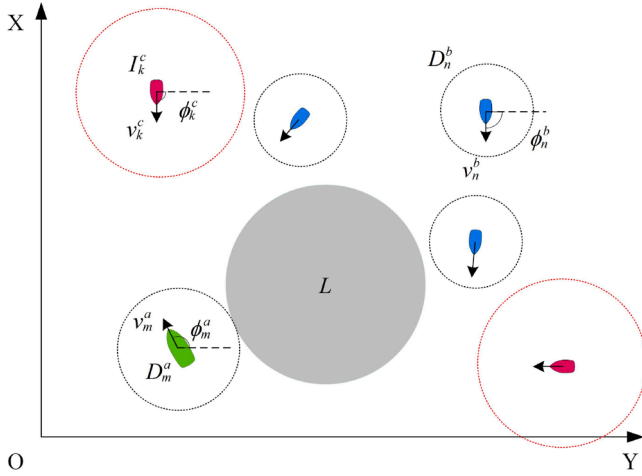


Fig. 1. Game of USVs around island.

applied to practice is discussed in combination with the designed hardware system. Finally, Section 5 summarizes the research findings of the paper, thoroughly discusses the theoretical and practical limitations of this work, and outlines future research directions.

2. Problem formulation

Consider a mission sea area without boundaries. The sea area contains two types of defensive USVs (defenders), D^a and D^b , multiple invasive USVs (targets) $I^c = \{I_1^c, \dots, I_i^c, \dots, I_M^c\}$, and one island L . The first type of defenders can be expressed as $D^a = \{D_1^a, \dots, D_m^a, \dots, D_M^a\}$; the second type of defenders can be expressed as $D^b = \{D_1^b, \dots, D_n^b, \dots, D_N^b\}$. Fig. 1 shows the plane geometry in the inertial Cartesian coordinate system. The area in the circle with a red dotted line represents the observation range of the target; the area in the circle with a black dotted line represents the capture range of the defender. v_m^a , v_n^b , and v_k^c represent the velocity of the first type of defender m , the velocity of the second type of defender n , and the velocity of the target k , respectively. ϕ_m^a , ϕ_n^b , and ϕ_k^c represent the heading of the first type of defender m , the second type of defender n , and the target k , respectively. In addition, M , N , and K represent the number of the three types of USVs mentioned above.

This study employs a widely used representative USV model in inertial Cartesian coordinates. The second-order kinematic equation can be expressed as:

$$\begin{cases} \dot{x}_i = v_i \cos(\phi_i) \\ \dot{y}_i = v_i \sin(\phi_i) \\ \dot{\phi}_i = \omega_i \end{cases}$$

由于实际中无法获取目标的角速度，我们将目标的最短入侵路径简化为目标与岛屿中心之间的连线。(1)

To further simplify the complexity of model construction and solution, this paper makes the following assumptions (Wang et al., 2024):

1. Different types of defenders are considered to have differences in parameters such as maximum velocity and capture radius.
2. The defender can obtain the situation information in the task sea area. However, the target has an observation radius and can only observe the situation information within this range.
3. If the target's action is entirely restricted, the target is considered incapacitated.
4. The area covered by the island is considered to be circular.

防御方是完全信息的假设！！！！

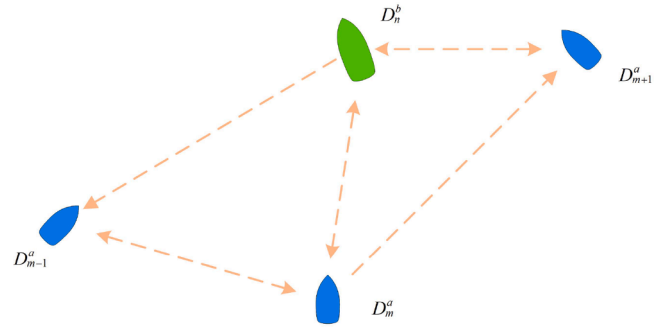


Fig. 2. Neighborhood topology in multi-USV systems.

3. Distributed game framework based on a hierarchical decision-making system

In multi-agent game scenarios, the strategic complexity grows exponentially with the number of participants, significantly complicating the analysis of inter-agent interactions and the mutual dependencies in their decision-making processes. In response to this problem, we introduced the concept of neighbor USVs. We define the other o defenders closest to defender a as its neighbor USVs, as shown in Fig. 2. In the decision-making process based on matrix game, USV will consider the possible strategy choices of neighbor USVs. Taking the USV D_m^a as an example, the arrows in the figure point to D_n^b from D_m^a and D_{m-1}^a , representing that D_n^b and D_{m-1}^a are the neighbors of D_m^a at the current moment. Due to the highly dynamic nature of the game process, the neighbor set of each USV undergoes continuous adaptation.

This paper proposes a distributed game framework based on a hierarchical decision-making system further. Under the framework, each defender makes decisions based on the obtained situation information. It is worth noting that in the decision-making process of the defender, the possible strategies of its neighbor USVs will be considered, and the equilibrium solution of the game will be sought on this basis (Li et al., 2022).

3.1. Guidance point prediction for dynamic defense

In the decision-making process of the defenders, finding an optimal equilibrium solution in a complex game environment to maximize its collaborative ability and effectively deal with the potential threat of targets is the core link. It is challenging to perform well only by paying attention to each participant's current motion information. The prediction of future strategies and status will solve this problem well. 决策时考虑对未来策略和状态的预测。

Preventing the targets from continuing to approach the island is the condition to win the game. Because the target's angular velocity cannot be obtained in practice, we simplify the target's shortest invasion path to the connection between the target and the center of the island. When the target moves along the shortest invasion path, its position can be expressed as Eq. (2). The nearest meeting point to the target is defined as the guidance point.

Where (x_k^c, y_k^c) denotes the target's position at the current moment, and $(x_k^{c'}, y_k^{c'})$ denotes the target's position at the next moment. Considering the underactuated characteristics of the USV, the distribution of the guidance points is shown in Fig. 3.

The red cone area represents the threat area of the target at the current moment, and is also the area where the shortest invasion path is located. The red line represents the shortest invasion path, ϕ_k^c represents the expected heading of the target as it moves along this path. The blue and green lines represent the paths that the defenders m and n need to go through to reach the guidance points, respectively. r_m^a and r_n^b represent the turning radius of defender m and defender n , respectively. θ_m^a and θ_n^b represent the central angles corresponding to the path of defender m and defender n along the turning circle, respectively. p_{mk}^a and p_{nk}^b represent

论文假设目标会沿连接其当前位置与岛屿中心的直线移动，这是目标入侵的最短路径。蓝色和绿色有最小转弯角度的限制，所以蓝虚线和绿虚线才是两个防御者的实际拦截路线，故蓝绿色点是防御方与目标预计相遇的最早位置点，也即引导点。

注意！！对于目标的位置预测的假设很粗糙，按理说目标也该有最小转角限制。

θ 代表了防御方为了拦截目标当前需要转过的角度，它是结合了USV自身的物理限制（转弯半径）与瞬时战场态势计算出的决策参数。而 r 是防御方固有的最小转弯半径。

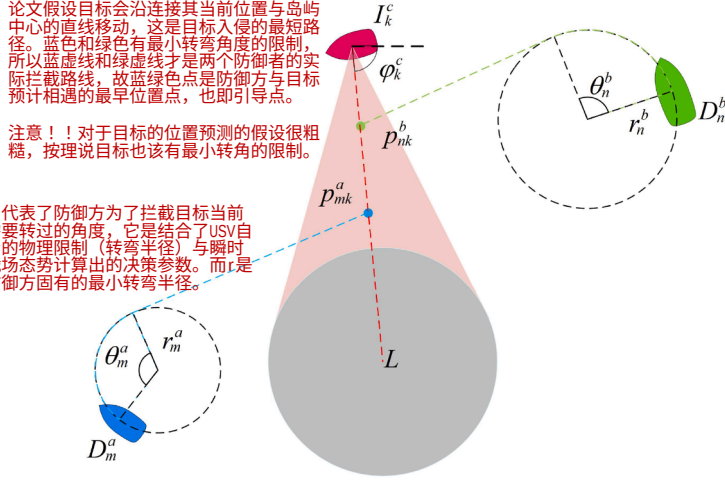


Fig. 3. Distribution of the guidance points.

the guidance points of defender m and defender n , respectively. Taking the defender D_m^a as an example, we establish the following equations to solve the two-dimensional position coordinates of the guidance point (Chen et al., 2024). Where, Δt is the total time of movement.

时间间隔 Δt 后对目标的位置坐标预测：

$$\begin{cases} (x_k^c, y_k^c) \begin{cases} x_k^c = x_k^c + v_k^c \cdot \cos(\varphi_k^c) \cdot \Delta t, \\ y_k^c = y_k^c + v_k^c \cdot \sin(\varphi_k^c) \cdot \Delta t \end{cases} \end{cases} \quad (2)$$

时间间隔 Δt 后USV向左转弯时的引导点求解：

$$\begin{cases} x_m^a + r_m^a \cdot (\cos(\phi_m^a - \pi/2) + \cos(3 \cdot \pi/2 - \theta_m^a + \phi_m^a)) \\ + (v_m^a \cdot \Delta t - \theta_m^a \cdot r_m^a) \cdot \cos(\phi_m^a - \theta_m^a) \\ = x_k^c + v_k^c \cdot \cos(\varphi_k^c) \cdot \Delta t \\ y_m^a + r_m^a \cdot (\sin(\phi_m^a - \pi/2) + \sin(3 \cdot \pi/2 - \theta_m^a + \phi_m^a)) \\ + (v_m^a \cdot \Delta t - \theta_m^a \cdot r_m^a) \cdot \sin(\phi_m^a - \theta_m^a) \\ = y_k^c + v_k^c \cdot \sin(\varphi_k^c) \cdot \Delta t \\ \Delta t \geq 0 \\ \theta_k^c \geq 0 \end{cases} \quad (3)$$

横线部分就是 (2) 中的目标坐标

时间间隔 Δt 后USV向右转弯时的引导点求解：

$$\begin{cases} x_m^a - r_m^a \cdot (\cos(\phi_m^a - \pi/2) + \cos(3 \cdot \pi/2 - \theta_m^a - \phi_m^a)) \\ + (v_m^a \cdot \Delta t - \theta_m^a \cdot r_m^a) \cdot \cos(\phi_m^a + \theta_m^a) \\ = x_k^c + v_k^c \cdot \cos(\varphi_k^c) \cdot \Delta t \\ y_m^a - r_m^a \cdot (\sin(\phi_m^a - \pi/2) - \sin(3 \cdot \pi/2 - \theta_m^a - \phi_m^a)) \\ + (v_m^a \cdot \Delta t - \theta_m^a \cdot r_m^a) \cdot \sin(\phi_m^a + \theta_m^a) \\ = y_k^c + v_k^c \cdot \sin(\varphi_k^c) \cdot \Delta t \\ \Delta t \geq 0 \\ \theta_k^c \geq 0 \end{cases} \quad (4)$$

如果 (3) (4) 方程组无解，说明USV无法在 Δt 内拦截目标，需重新分配任务。

By introducing the solution of the above equations into Eq. (2), the position of the guidance point can be obtained, which can be used as the basis for evaluating the ability of the defender to prevent the intrusion of the specific target. It should be noted that the solutions of the above equations do not always exist. When the above equations have no solution, the defender may be unable to prevent the target. In addition, How to select the strategy to improve the defenders' coordination ability on the basis of ensuring real-time is also a problem to be solved.

Aiming at the problem of strategy space dimension disaster caused by continuous action space in the game, this study innovatively proposes a task-action decoupling hierarchical decision-making framework based on game theory. The framework adopts a two-layer optimization architecture design: based on the Nash equilibrium solution, the upper layer realizes optimal task allocation, and the lower layer completes fine

action strategy generation. This hierarchical architecture decouples the high-dimensional decision space into two low-dimensional subspaces. It combines the task characteristics to constrain the optional strategy set of participants, which significantly improves the scalability of decision-making methods in large-scale collaborative scenarios to ensure that the output conforms to the participants' dynamic constraints.

3.2. Upper game model

This paper uses a four-tuple $\langle P_U, S_U, A_U, R_U \rangle$ to describe the upper game model (Kartik and Nayyar, 2021). P_U is the set of participants in the game. Here, we take the defender as a participant in the game and its neighbor USVs as another participant in the game. S_U is the position, velocity, and heading information of all defenders and all targets. A_U represents the participants' upper strategies. A defender's strategy set can be expressed as $a = \{a_1, \dots, a_k, \dots, a_K\}$, where a_k represents the defender to prevent the target K . R_U represents the payoff function of the participants, as shown below:

$$R_U = \omega_1 \cdot r_1 + \omega_2 \cdot r_2 + \omega_3 \cdot r_3 \quad (5)$$

Among them, r_1 , r_2 and r_3 represent the benefit of the task, the cost of resource consumption and the cost of potential risks, respectively. ω_1 , ω_2 and ω_3 represent the weight coefficients. r_1 can be expressed as:

$$r_1 = \sum_{k=1}^K \left(\left(\sum_{m=1}^M (B_{mk}^a \cdot M P_{mk}^a) + \sum_{n=1}^N (B_{nk}^b \cdot M P_{nk}^b) \right) \right) / z_1 \quad (6)$$

任务完成的收益

Where B^a , B^b , $M P^a$, $M P^b$ are binary decision matrices. When B_{mk}^a is equal to 1, defender m adopts an upper strategy to prevent target k . When B_{mk}^a is equal to 0, defender m does not prevent target k . B^b has the same function as B^a . $M P_{mk}^a$ is used to determine whether there is a guidance point between defender m and target k . If $M P_{mk}^a$ is equal to 0, the guidance point does not exist; otherwise, the guidance point exists. $M P^b$ has the same function as $M P^a$. z_1 is a positive integer.

r_2 can be expressed as: 资源消耗的成本

$$r_2 = \sum_{k=1}^K \left(\sum_{m=1}^M \left(\sqrt{(x_m^a - x_{M_{mk}}^a)^2 + (y_m^a - y_{M_{mk}}^a)^2} \cdot B_{mk}^a \right) + \sum_{n=1}^N \left(\sqrt{(x_n^b - x_{M_{nk}}^b)^2 + (y_n^b - y_{M_{nk}}^b)^2} \cdot B_{nk}^b \right) \right) \quad (7)$$

Where $(x_{M_{mk}}^a, y_{M_{mk}}^a)$ is the coordinates of the guidance point between the target k and the defender m . $(x_{M_{nk}}^b, y_{M_{nk}}^b)$ is the coordinates of the guidance point between the target k and the defender n . When the guidance point does not exist, $(x_{M_{mk}}^a, y_{M_{mk}}^a)$ or $(x_{M_{nk}}^b, y_{M_{nk}}^b)$ is the current position of the target k .

r_3 can be expressed as:

$$r_3 = \sum_{k=1}^K r_3^k \quad (8)$$

潜在(未成功拦截)的风险成本

$$r_3^k = z_2 / \left(\left(\sum_{m=1}^M (B_{mk}^a \cdot M P_{mk}^a \cdot c r_m^a) + \sum_{n=1}^N (B_{nk}^b \cdot M P_{nk}^b \cdot c r_n^b) \right) + z_3 \right) \quad (9)$$

Where z_2 and z_3 are positive numbers, and z_2 is much larger than z_3 . $c r_m^a$ and $c r_n^b$ denote the capture radius of the defender m and the defender n .

3.3. Lower game model

The lower game is also described by four tuples $\langle P_L, S_L, A_L, R_L \rangle$, where P_L and S_L are the same as P_U and S_U in the upper level game, respectively. However, the lower game focuses more on the defender's basic maneuvers. Therefore, the set of strategies A_L available to each

4(a)的核心目的是通过协同布防组织目标沿最短入侵路径进入受保护区域。红色锥形区域是目标的威胁角，代表目标可能入侵的范围；橙色和蓝色是由防御者形成的总覆盖角度。需要防御者的覆盖角度完全覆盖目标的威胁角才能获胜。

4(b)则是通过形成包围圈对目标进行捕获，形成规则：至少3个防御者处于目标的捕获半径内，且相邻防御者夹角不超过180度。

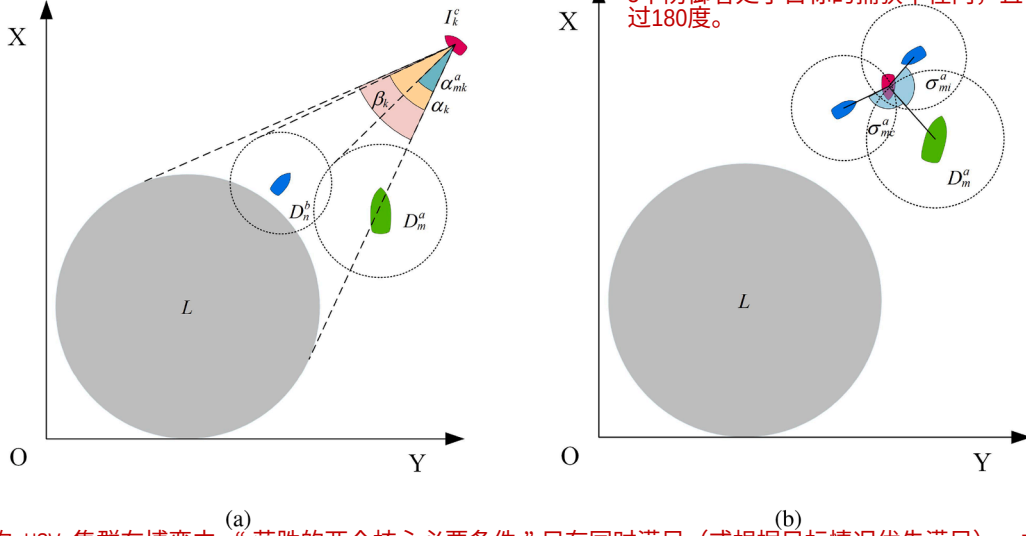


图 4

图 4. Necessary conditions for winning in non-cooperative games. (a) Intercepting the target (b) Capturing the target.

defender includes constant left turn, constant execution, constant right turn, accelerating left turn, accelerating straight, accelerating right turn, decelerating left turn, decelerating straight, decelerating right turn. The payoff function R_L for the lower game is as follows:

$$R_L = \omega_4 \cdot r_4 + \omega_5 \cdot r_5 + \omega_6 \cdot r_6 \quad (10)$$

Among them, r_4 , r_5 and r_6 represent the benefit of the task, the cost of resource consumption and the cost of potential risks, respectively. w_4 , w_5 and w_6 represent the weight coefficients. r_4 can be expressed as:

两类防御者机动前后距离引导点的距离，靠近的话(12-13)是负数

$$r_3 = \sum_{k=1}^M \left(\sum_{m=1}^M (r_3^{mk} \cdot B_{mk}^a) + \sum_{n=1}^N (r_3^{nk} \cdot B_{nk}^b) \right) \quad (11)$$

虽然越靠近引导点，负值越大，但是对应的协同项奖励也越大，所以相对的RL还是更大。

$$r_3^{mk} = \begin{cases} \sqrt{(x_m^{a'} - x_{Mmk}^a)^2 + (y_m^{a'} - y_{Mmk}^a)^2} \\ -\sqrt{(x_m^a - x_{Mmk}^a)^2 + (y_m^a - y_{Mmk}^a)^2} & , d_{mk} \geq CR_m^a \\ 0 & , d_{mk} < CR_m^a \end{cases} \quad (12)$$

$$r_3^{nk} = \begin{cases} \sqrt{(x_n^{b'} - x_{Mnk}^b)^2 + (y_n^{b'} - y_{Mnk}^b)^2} \\ -\sqrt{(x_n^b - x_{Mnk}^b)^2 + (y_n^b - y_{Mnk}^b)^2} & , d_{nk} \geq CR_n^b \\ 0 & , d_{nk} < CR_n^b \end{cases} \quad (13)$$

Here, $(x_m^{a'}, y_m^{a'})$ and $(x_n^{b'}, y_n^{b'})$ denote the position of defender m and defender n at the next moment after they adopt the current strategy respectively. d_{mk} and d_{nk} represent the distance between target k and defenders m and n , respectively. CR_m^a and CR_n^b denote the capture radius of defenders m and n , respectively.

The benefit of collectivity is more focused on enhancing the ability of cooperation between different defenders to achieve better performance in the game, as shown in Fig. 4.

Fig. 4(a) shows the first necessary condition for winning in the game, that is, the interception of the target. Condition 1 requires that any target with mobility cannot enter the protected area before the end of the round. This condition holds when the defender's safety angle α_k completely covers the target's threat angle β_k . Where β_k denotes the opening angle of the target's conical threat zone, and α_k denotes the defender's safety angle, the range of which must always cover β_k to ensure successful interception.

Fig. 4(b) shows the second necessary condition, that is, the capture of the target. This condition requires at least three defenders to be located within the capture radius of the target, and the angle between any

adjacent defenders should not exceed 180° . From a geometry perspective, this constraint ensures a stable triangular bounding structure, so the other two can quickly fill the vacancies left by any defender. The 180° limit is critical to maintaining the encirclement efficiency and stability. If the angle exceeds 180° , the defender is located on the same side of the target, which will seriously weaken its ability to limit the target's movement.

In view of the above two necessary conditions, we divide the benefit of the collectivity r_5 into interception part and capture part. r_5 can be expressed as: 形成包围圈的奖励，主要从角度层面

$$r_5 = w_7 \cdot \sum_{m=1}^M \left((\sigma_{mc} - 2\pi/N_m^a)^2 + (\sigma_{mi} - 2\pi/N_m^a)^2 \right) + w_8 \cdot \sum_{k=1}^K (\alpha_k / \beta_k) \quad (14)$$

Where σ_{mc}^a and σ_{mi}^a denote the angles between defender m and the defender in its clockwise direction and the angles between defender m and the defender in its counterclockwise direction, respectively.

Because many factors must be considered in obstacle avoidance, we divide the benefits r_6 into maintaining a safe distance from members within the group r_6^a , a safe distance from targets r_6^b , and a safe distance from the island r_6^c . r_6 can be expressed as (Yan et al., 2021):

$$r_6 = r_6^a + r_6^b + r_6^c \quad (15)$$

避障安全收益：避免防御者之间、防御者与目标、防御者与岛屿碰撞。

r_6^a , r_6^b and r_6^c can be expressed as:

$$r_6^a = \sum_{u=1}^U \sum_{u'=u+1}^U r_{6,uu'}^a \quad (16)$$

$$r_6^b = \sum_{u=1}^U \sum_{k=1}^K r_{6,uk}^b \quad (17)$$

$$r_6^c = \sum_{u=1}^U r_{6,uL}^c \quad (18)$$

$$r_{6,uu'}^a = \begin{cases} 1, & d_{uu'} \geq D_2 \\ (d_{uu'} - D_1) / (D_2 - D_1) & D_1 < d_{uu'} < D_2 \\ 0 & d_{uu'} \leq D_1 \end{cases} \quad (19)$$

$$r_{6,uk}^b = \begin{cases} 1, & d_{uk} \geq D_2 \\ (d_{uk} - D_1) / (D_2 - D_1) & D_1 < d_{uk} < D_2 \\ 0 & d_{uk} \leq D_1 \end{cases} \quad (20)$$

$$r_{6,uL}^c = \begin{cases} 1, & d_{uL} \geq D_3 + r_L \\ (d_{uL} - r_L)/D_3 & r_L < d_{uL} < D_3 + r_L \\ 0 & d_{uL} \leq r_L \end{cases} \quad (21)$$

Where D_1 , D_2 , and D_3 are constants; $d_{uu'} = \sqrt{(x_u - x_{u'})^2 + (y_u - y_{u'})^2}$ denotes the distance between two defenders; $d_{uk} = \sqrt{(x_u - x_k^c)^2 + (y_u - y_k^c)^2}$ denotes the distance between the defender and the target k ; $d_{uL} = \sqrt{(x_u - x_L)^2 + (y_u - y_L)^2}$ represents the distance between the defender and the island. (x_u, y_u) and $(x_{u'}, y_{u'})$ represent the position coordinates of the defenders u and u' in the set of the current defender and its neighbor USVs. U denotes the number of elements contained in the above set. (x_L, y_L) denotes the coordinates of the center of the island, r_L denotes the radius of the island area. The pseudocode of the proposed strategy is described in Algorithm 1.

Algorithm 1: The distributed cooperative strategy.

```

1 Input: Position, velocity, heading and maneuverability of each
  USV.
2 Output: The angular velocity and acceleration of each defender.
3 Initialization allocation scheme.
4 for each optional strategy  $a_i \in A_U$  of the defender in the upper
  game do
5   Calculate guidance point between defender and each
   target using Eqs. (3) and (4).
6   for each optional strategy combination  $a_{-i} \in A_U$  of neighbor
   defenders do
7     Calculate payoff  $R_U(a_i, a_{-i})$  using Eq. (5).
8   end
9 end
10 Solve upper game Nash equilibrium.
11 Assign optimal target for each defender.
12 for each optional maneuver strategy  $a_i \in A_L$  of the defender in the
  lower game do
13   for each optional strategy combination  $a_{-i} \in A_L$  of neighbor
   defenders do
14     Calculate payoff  $R_L(a_i, a_{-i})$  using Eq. (10).
15   end
16 end
17 Solve lower game Nash equilibrium.
18 Determine optimal maneuver (angular velocity and
  acceleration) for each defender

```

这个算法完全就是机械的计算上下层，分别得到谁该拦截谁以及机动控制策略，甚至连均衡的计算方式也没说，只是说计算每一层的纳什均衡（怀疑是遍历的方式）。

4. Simulation experiment

In order to systematically verify the advantages of the proposed algorithm (PA) in this paper, the simulation experiment part is designed as three key verification modules: algorithm feasibility verification, underactuated characteristic impact analysis, and distributed architecture scalability test based on hierarchical decision-making. This modular experimental design not only corresponds to the core innovation of this paper, but also comprehensively demonstrates the comprehensive performance advantages of the proposed algorithm through multi-dimensional comparative experiments with the traditional centralized method (CA), the method that does not predict the future state of participants (NP), and the method that does not consider the underactuated constraint in the prediction process (NUA) (Li et al., 2019). Reinforcement learning methods were excluded from comparisons due to their data and resource requirements, which are incompatible with our experimental conditions. Instead, we focus on online decision-making methods under comparable resource constraints. Through quantitative index analysis

Table 1
Basic parameters of simulation.

Parameter type	Parameter value
The maximum velocity of the first-class defender	10 m/s
The maximum angular velocity of the first-class defender	0.2 rad/s
The maximum acceleration of the first-class defender	2 m/s ²
The capture radius of the first-class defender	200 m
The maximum velocity of the second-class defender	8 m/s
The maximum angular velocity of the second-class defender	0.1 rad/s
The maximum acceleration of the second-class defender	1 m/s ²
The capture radius of the second-class defender	300 m
The observation distance of the target	500 m
The radius of the island	300 m

and typical scene visualization, the experimental conclusions are persuasive. We consider a game scenario for a heterogeneous USV cluster containing two types of defenders: three first-class defenders and one second-class defender. Compared with the second type, the first type has better motion performance, manifested in its faster velocity and angular velocity. The second type of defender has a larger capture radius. The basic parameters of the simulation are set as shown in Table 1.

It should be added that at the beginning of the mission, the defenders are randomly distributed in a circular area within 1000 m from the island's center. The targets are randomly distributed in an area 1500 m away from the island. In addition, in the upper game payoff function, the weights are set as $w_1 = 0.6$, $w_2 = 0.2$, and $w_3 = 0.2$. This configuration is based on the following rationale: a higher w_1 ensures that the defender prioritizes meeting the essential condition 2, while appropriately set w_2 and w_3 guarantee that, in the absence of a guidance point, the defender can effectively steer toward the essential condition 1. The weights for the lower game payoff function are set as $w_4 = 0.2$, $w_5 = 0.2$, and $w_6 = 0.6$. This parameter setup primarily aims to balance individual and collective interests while emphasizing collision avoidance as the top priority. The safety distance parameters are determined mainly to give participants sufficient time for strategic adjustments. Considering a maximum relative speed of 20 m/s, the dynamic collision avoidance distances D_1 and D_2 are set to 60 m. For static obstacle collision risks, D_3 is conservatively set to 30 m.

4.1. Algorithm feasibility verification

This chapter verifies the proposed method's basic effectiveness in a complex dynamic environment through typical game scenario tests. Considering that the target's motion performance may be higher than the defender in the actual situation, we test and analyze the two situations in which the targets are at a disadvantage and at least one target is at an advantage.

4.1.1. Targets are at a disadvantage

We conducted 200 random simulation experiments, and the maximum game time per round was set to 600 s. During the experiment, the velocity of the target was randomly distributed between 6 m/s and 8 m/s. The game process diagrams are as follows. In order to show the trajectory of each defender and each target in the game process more clearly, we split the whole game process into three stages.

In the Fig. 5, three figures of each row represent the trajectories of each USV in the same round. The blue maker represents the first type of defender, and a blue circle represents its capture range; a green maker identifies the second type of defender, and a green circle represents its capture range; a red maker marks the target before it is not captured, and the red maker will become black after losing the ability to move. In the above game process, the velocity and heading change curves of each defender and each target are as follows.

It can be seen from Fig. 6 that when the targets are all at a disadvantage, the defender can win the game by capturing all the targets, as described in Condition 2. Taking round 1 as an example, the speeds of

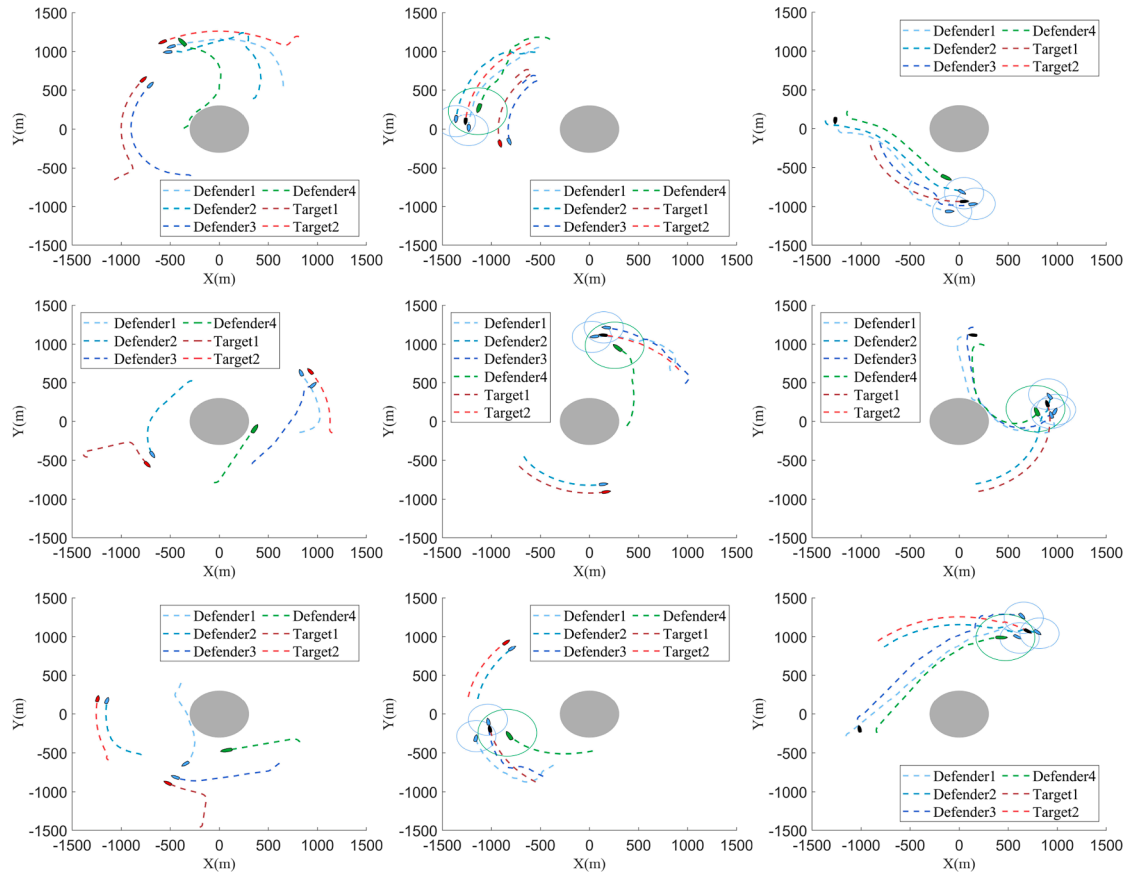


Fig. 5. Trajectory of each USV when the targets are at a disadvantage.

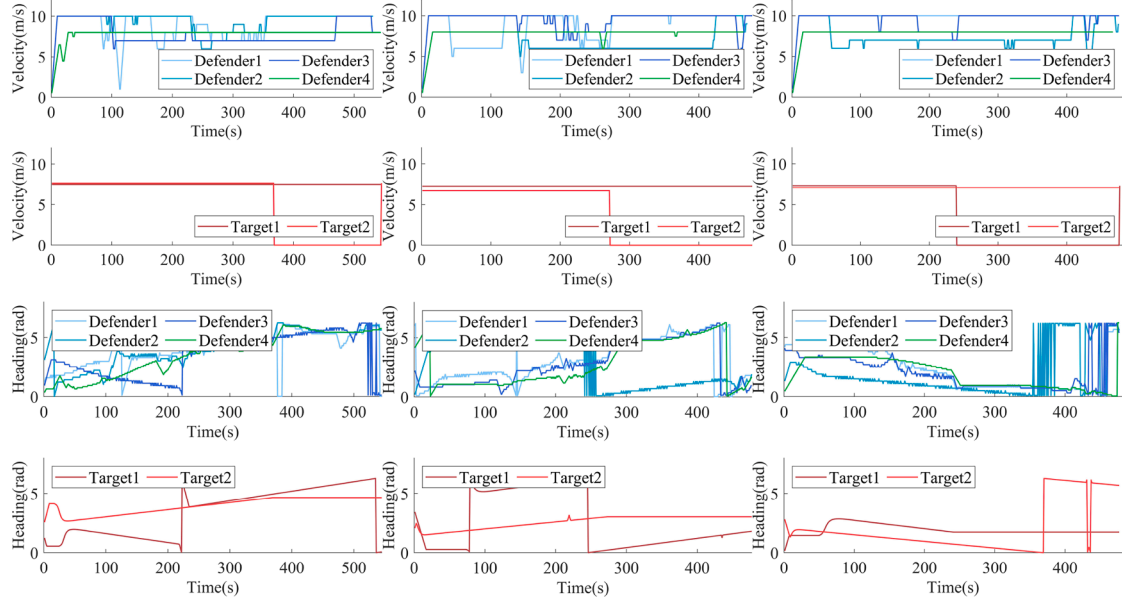


Fig. 6. Velocity and heading change curve when the targets are at a disadvantage.

target 1 and target 2 are 7.6 m/s and 7.7 m/s, respectively. At the beginning of the task, due to the difference in position distribution and motion parameters between the two sides of the game, defenders 1, 2, and 4 first approach target 2 and gradually expand the capture trend. At the same time, defender 3 is close to target 1, preventing target 1 from continuing to approach the island. Subsequently, defenders 1, 2, and 4 completed the capture of target 2 at 368 s and turned closer to target 1. Finally, the defenders 1, 2, and 3 captured target 1 at 545 s. At this

time, all targets lose mobility, and the game is completed. However, this method of winning the game is not always successful, which is greatly affected by the location distribution of both sides.

4.1.2. At least one target is at an advantage

To further explore the proposed strategy's ability to cope with superior targets, we set the target's maximum velocity to 12 m/s. The rest of the settings are the same as in Section 5.1.

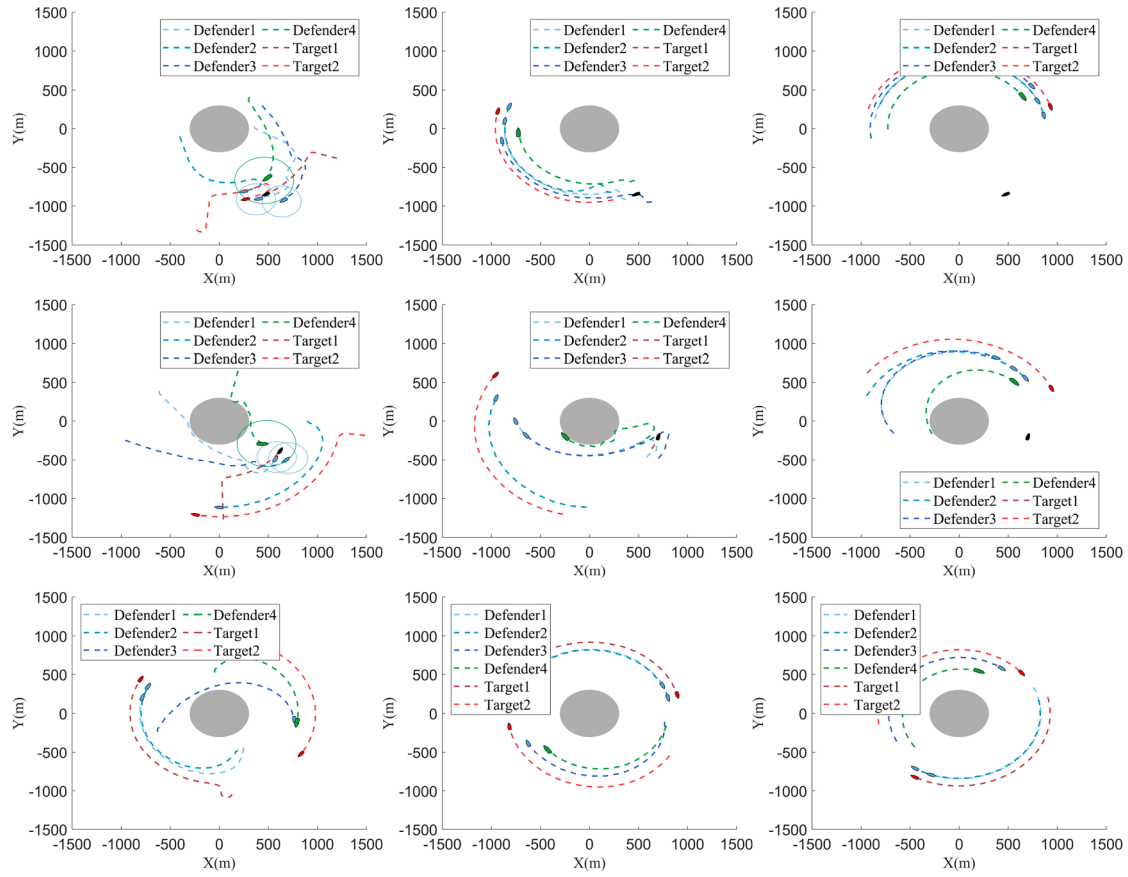


Fig. 7. Trajectory of each USV when at least one target is at an advantage.

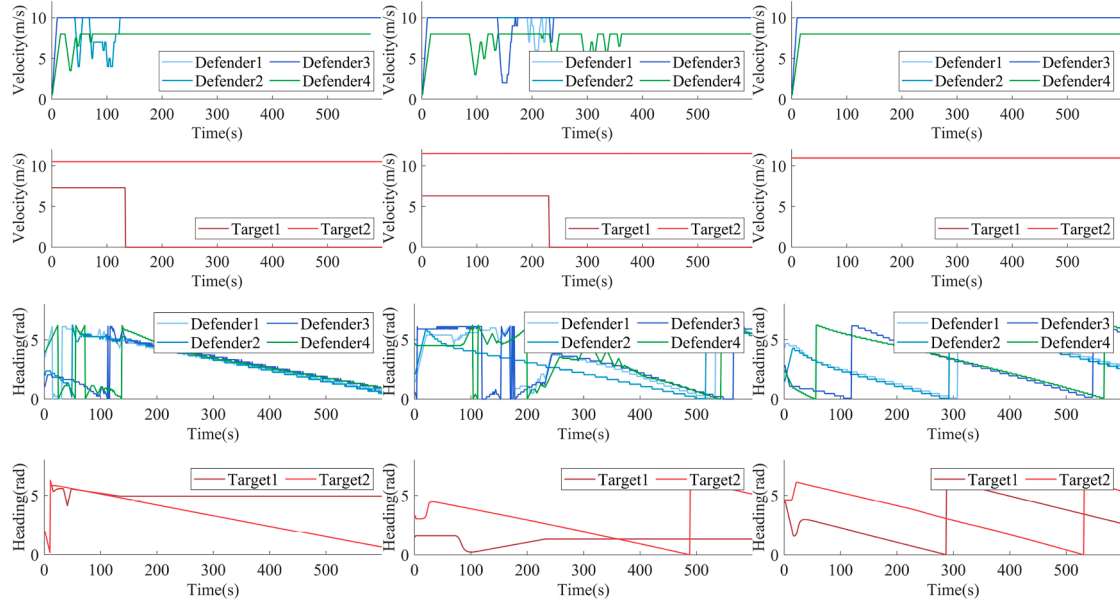


Fig. 8. Velocity and heading change curve when at least one target is at an advantage.

In the above game process, the velocity and heading change curves of each defender and each target are as follows.

Figs. 7 and 8 show that when only one advantage target exists, the disadvantage target will be preferentially captured. Then, the defender concentrates on preventing the dominant target until the end of the simulation. The speed of target 1 in round 1 and round 2 is 7.3 m/s and 6.4 m/s respectively, both of which are inferior targets. They are

captured by defenders 1,3,4 at 135 s and 228 s after the start of the round.

In round 3, the speed of targets 1 and 2 exceeds 10 m/s. Both of these targets are in a dominant position and are not likely to be captured by defenders. Therefore, the defender will try to delay the time required for the target intrusion, as described in Condition 1. Defenders 1 and 2 block target 1, while defenders 3 and 4 block target 2. Throughout the

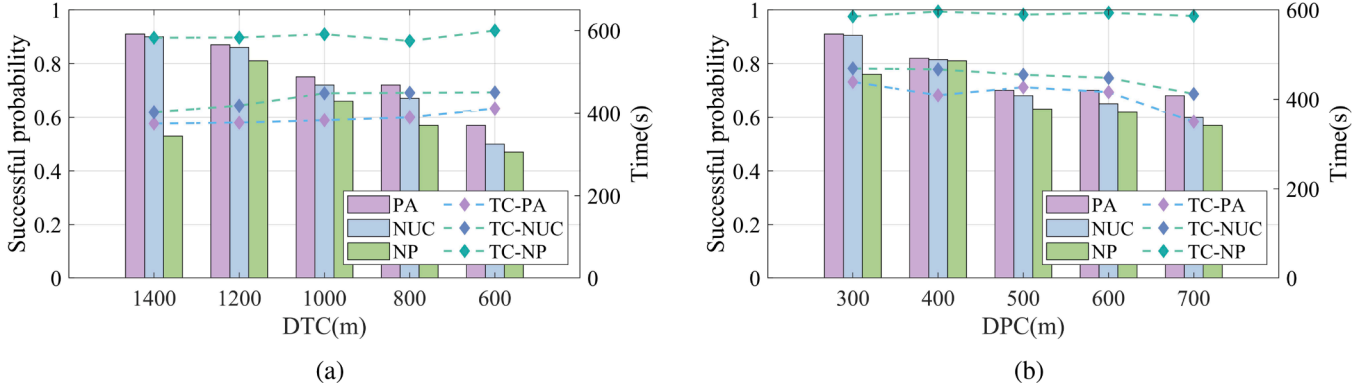


Fig. 9. Impact analysis of underactuated characteristics using Monte Carlo simulation. (a) DTC (b) DPC.

game, each defender's speed and heading changes are within a reasonable range.

It is worth noting that when the advantage of the target is significant, the target can reach the island as the game time increases. In round 1, defenders 1, 3, and 4 completed the capture of the inferior target 1 shortly after the simulation began and then, together with defender 2, prevented the superior target from approaching the island. Since target 2 is at an advantage, the defenders have been unable to expand the capture trend but move on the circular route closer to the island. Finally, target 2 failed to reach the island before the end of the simulation. Although the method proposed in this paper cannot guarantee the game's success in all cases, it can effectively extend the time required for successful intrusion even in a failed round, which will be verified in more detail in the Monte Carlo simulation. In addition, the distribution of the initial positions of the two sides also dramatically affects the process and results of the game. These factors will be discussed as independent variables in Monte Carlo simulation.

4.2. Underactuated characteristic impact analysis

This section focuses on verifying the improvement effect of decision quality after introducing underactuated constraints in reachable set analysis, and compares PA with NP and NUA to quantify its superiority (Jensen et al., 2014). We design two sets of Monte Carlo simulations based on the distance from the patrol route to the center of the protected area (DPC) and the distance from the target to the center of the protected area (DTC) to reflect the difference in decision quality of different methods effectively. The simulation results are shown in Fig. 9.

Fig. 9(a) shows the Monte Carlo simulation results when DPC is 500 m and DTC is 1400 m, 1200 m, 1000 m, 800 m, and 600 m, respectively. Fig. 9(b) shows the Monte Carlo simulation results when DTC is 1000 m, and DPC is 300 m, 400 m, 500 m, 600 m, and 700 m, respectively. It can be seen from the figure that the two indicators of task success probability and successful round consumption time corresponding to NP and NUA are not as good as the PA. This is mainly because the accurate solution of the guidance point can make the defender reach the target faster and affect the target's action. This also means the target will be blocked farther away from the protected area. As the difference between DPC and DTC decreases, the success rate of the PA algorithm gradually decreases, but it is still higher than that of NP and NUA. At the same time, the gap between the PA algorithm and the NUA is gradually increasing. This is due to the gradual decrease of DPC and DTC, which leads to the decrease of the initial distance between the defender and the target. The defender needs more accurate guidance to achieve better performance near the target. In addition, the success probability corresponding to the NP increases first and then decreases, which also shows that the robustness of the NP is weaker than that of the PA.

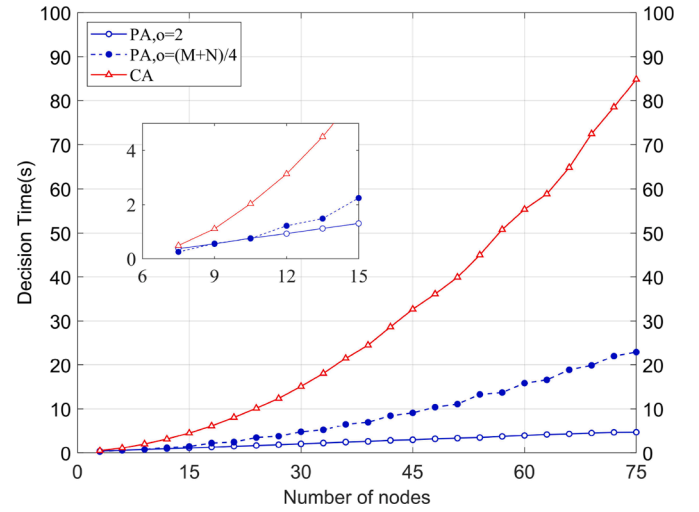


Fig. 10. Comparison of decision time.

4.3. Scalability verification of distributed hierarchical decision-making architectures

4.3.1. Scalability verification of decision time

This section systematically evaluates the proposed hierarchical distributed decision architecture's performance in ensuring real-time decision-making for task scenarios with different complexities. Fig. 10 shows the comparative simulation results of PA and CA in decision time when the number of neighbor nodes o takes different values.

The decision time of this architecture mainly depends on the solution of the guidance point and the Nash equilibrium, which is essentially affected by the number of neighboring nodes and the number of targets. The experimental results show that under the same hardware conditions, compared with CA, the decision-making time of the framework proposed in this study is more gentle with the growth trend of cluster size, reflecting good scalability. When the total number of nodes is about 20, the single decision-making time is about 1s, which has the potential to be applied to engineering practice. Compared with the traditional centralized decision-making method, this framework can significantly reduce the computational burden of a single node through task-action decoupling and distributed computing. When the number of nodes is large in the practical application process, the scale of neighbor USVs can be adjusted to balance the decision-making time and quality.

4.3.2. Scalability verification of decision quality

This section systematically evaluates the performance of the proposed hierarchical distributed decision-making architecture in ensuring

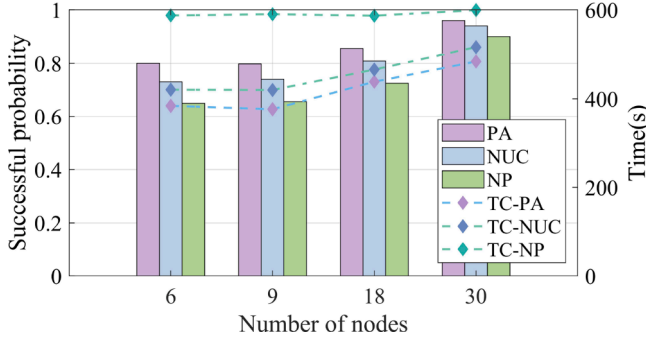


Fig. 11. Comparison of decision quality.

decision quality for the task scenarios with a total number of nodes of 6, 9, 18, and 30. Fig. 11 shows the comparative simulation results of PA, NUC, and NP when the number of neighbor nodes is 2.

The experimental results show that under the same task scenario, the successful probability shows a monotonically increasing trend as the node scale expands from low to high density. However, at the same time, the average time consumption of successful rounds also increases synchronously. The following mechanism can explain this seemingly contradictory phenomenon: although the increase in the number of nodes will inevitably increase the complexity of system collaborative decision-making, the spatial coverage advantage brought by high-density node deployment is gradually highlighted due to the constant key scene parameters, such as the size of the protected area. Specifically, in the dynamic task environment, the increase in node density significantly enhances the regional coverage ability, which not only makes up for the efficiency loss caused by the increase in decision complexity but also ultimately improves the overall task completion probability.

It is worth noting that when the number of nodes exceeds 18, the time spent on successful rounds increases significantly, indicating an increase in the proportion of tasks terminated due to the maximum round time. In particular, when the total number of nodes reaches 30, the advantages of the PA algorithm are weakened compared with the NUC and NP strategies. This is mainly because the PA algorithm requires more accurate prediction when performing guidance point calculation, and the increase in the number of targets in the task leads to longer decision time and lower real-time performance. In the dynamic game task, this deci-

Table 2

Hardware system configuration of the prototype platform.

Module category	Key components
Power supply module	Lithium battery
Main controller	Autopilot
Auxiliary controller	Raspberry Pi
Positioning module	Global Positioning System (GPS)
Execution module	Rudders and brushed motors
Communication module	Serial Communication Module AS69-T20

sion delay will directly affect the decision efficiency, thus limiting the performance improvement of the PA algorithm in the high node number scenario.

In practical applications, to further expand the scale of nodes and maintain the superior performance of the PA algorithm, it is necessary to adjust the algorithm adaptively. Specifically, when predicting the guidance point, the target can be screened according to the distance factor or the threat level, and the low-priority target can be eliminated to reduce the computational load, thereby improving the decision-making efficiency.

4.4. Practical implementation considerations

This study developed a hardware system, as shown in Fig. 12, and the key components of each module are detailed in Table 2. The system is modularized into the power supply, control, positioning, execution, and communication units. The selection of key components balances performance, cost, and reliability in the marine environment. The designed hardware system, working with the host computer, gives the theoretical method proposed in this study practical application potential.

The distributed decision-making method proposed in this study can significantly improve the system's real-time response performance by optimizing the calculation process and decision-making mechanism. This advantage ensures that the quality of algorithm decision-making meets the practical application standards and shows outstanding collision risk avoidance ability in multi-USV collaborative operation, which lays an essential foundation for its engineering application.

It should be pointed out that the current research is based on the idealized assumption that a single USV can obtain global observation information. This assumption is feasible in small-scale test scenarios but may have limitations in practical large-scale mission sea area

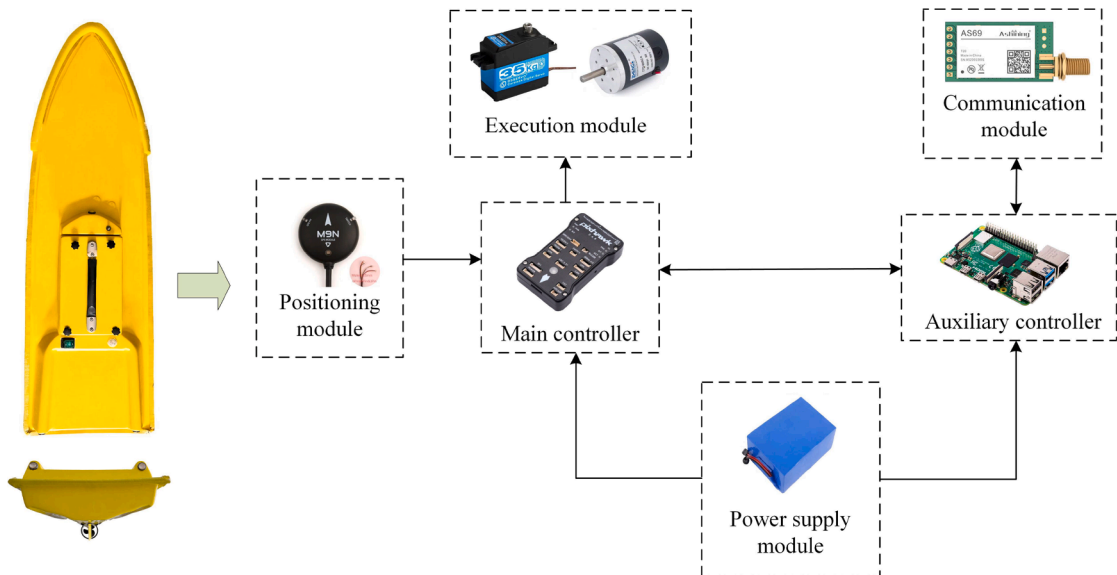


Fig. 12. Schematic of the prototype platform and hardware system.

applications. Therefore, it is suggested that a high-performance communication networking method be developed as a supplement to achieve accurate control of communication objects, communication frequency, and communication content between USVs. These technical supplements will effectively enhance distributed systems' operational reliability and scale scalability in complex marine environments.

5. Conclusion

The computational burden of the heterogeneous USV cooperative game algorithm proposed in this study mainly focuses on two key links: guidance point prediction and Nash equilibrium solution. Among them, the accurate solution of the guidance point plays a decisive role in task allocation and maneuver strategy optimization. However, with the expansion of the cluster size, the solution time of the guide point shows a linear growth trend, which directly affects the real-time performance of the algorithm. In order to solve this problem, this study innovatively adopts a hierarchical decision-making framework and a distributed computing framework, effectively reducing the computing load of a single node. The experimental results show that the optimization strategy significantly improves the task success probability and reduces the task completion time while ensuring real-time decision-making.

It is worth noting that the method proposed in this study can be further extended to the hybrid cluster collaboration scenario composed of manned / unmanned surface vehicles. Compared with the USVs cluster, this scenario differs mainly in the constraints of communication conditions: when reliable communication cannot be established, the problem will evolve into a more challenging three-party game problem, which will be one of the key directions of follow-up research.

It should be pointed out that some aspects need to be improved in this study. The current research assumes that the defender can obtain completely accurate environmental information, including position, speed, heading, etc., which is different from the actual application. The next step will focus on the game problem under partially observable conditions. In addition, the existing algorithms are based on the deterministic prediction of the future state. At the same time, the sensor accuracy and environmental interference in the actual environment will lead to the interval characteristics of the information. Although the distributed architecture can still maintain sub-second response within the scale of 20 nodes, it is necessary to develop a more robust situation awareness algorithm to further improve the quality of decision-making.

CRedit authorship contribution statement

Cong Chen: Writing – review & editing, Software, Methodology, Writing – original draft, Resources; **Xiao Liang:** Software, Methodology, Project administration, Conceptualization; **Zhao Zhang:** Software, Conceptualization; **Dianying Liu:** Validation, Software; **Changdong Yu:** Validation, Software; **Wei Li:** Resources, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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